AstroPixel: Mission Designer: pitch script

Team Shay Rafe3 · NASA Space Apps 2026 · Space Mission Design Game · Ramallah · 5-minute slot: 3:40 presentation (3:00 spoken + 0:40 video), then Q&A · Science checked against the team's Science Study Guide, Scientific Audit and Judges' Preparation (14 Nov 2026)

The story in one line (everyone memorises it):
"Millions love space, but few tools let them run a whole mission. AstroPixel lets anyone build and fly one, with real physics and NASA reference data, and learn why every part depends on every other."

1. Timing plan

#	Slide	Time	Length	Criteria	Speaker
1	Title: one line, then click	0:00–0:05	5 s	Presentation	
2	**The hook (the three numbers)** + the problem	0:05–0:50	45 s	Impact	
3	Close to home: Palestine	0:50–1:20	30 s	Impact	
4	Solution: AstroPixel	1:20–1:55	35 s	Creativity, Relevance	
5	Science + NASA data	1:55–2:30	35 s	Relevance, Validity	
6	Demo video (5 s intro + 40 s video)	2:30–3:15	45 s	Validity, Presentation	demo driver
7	Impact, playtest feedback, team + close	3:15–3:40	25 s	Impact, Validity	

The presentation takes **3:40** (3:00 spoken + 0:40 video), leaving about 1:20 for questions. Rehearse with a timer and stop at 3:40.

2. The script, slide by slide

Slide 1: Title `0:00–0:05`

> "We're Team Shay Rafe3 from Ramallah." *[click straight to slide 2]*

Slide 2: The hook + the problem `0:05–0:50`

> *[point at each number as you say it; slow down]*
> "**149 million** views of NASA's Artemis II coverage this year. **406 million** follows of NASA's accounts. **114,000** people signed up for Space Apps last year.
>
> The world loves space. But how many of those people have ever designed a mission themselves?
>
> Today they can **watch** a launch, but not the hundreds of decisions behind it. They can **study** one topic at a time: orbits, then power, then radios. Or they can **play** games that are mostly about flying, rarely about budget, power and data.
>
> **Few tools let you run a whole mission yourself, from the first idea to the final result.**"

The hook is the three numbers. Pause after "designed a mission themselves?" If asked: views and account follows are not unique people.

Slide 3: Close to home `0:50–1:20`

> "Here in Palestine, that gap is even wider. We found no aerospace engineering degree at our universities. The best-known way into NASA is a trip for a handful of teenagers. Our astronomy clubs do great work, but trying mission design yourself is hard.
>
> Young Palestinians are as curious about space as anyone. We want to add to what already exists.
>
> **The curiosity is here. Tools should be too.**"

Check before the pitch: the Palestine facts (section 6). Present them as what *we* found, not as a full survey.

Slide 4: Our solution `1:20–1:55`

> "So we built AstroPixel: a whole mission, in your browser.
>
> You pick one of 8 worlds and a task, and build your spacecraft from 29 parts based on real hardware while Mission Control tracks every resource live. You launch, cruise through events inspired by real missions and real 2024 solar storms, and land in each world's gravity. When something fails, the report tells you exactly why, so you fix it and fly again, alone or as a team of four roles.
>
> And players discover what engineers know: **add one camera, and mass, power, data and fuel all change. Sometimes you need a bigger battery or even a bigger rocket, and that costs money.**"

The banner numbers (Moon direct plan): the camera is 3 kg and 7 W and adds about 5 kg of fuel before reserve. Don't say every instrument *always* needs a bigger rocket: the existing array or rocket may absorb it.

Slide 5: The science + NASA data `1:55–2:30`

> "Underneath, it's real science. The game runs the rocket equation: on our Moon plan, a 1,000 kg ship needs about 1,700 kg of fuel. Sunlight fades with distance squared: Mars gets 43% of Earth's, Saturn about 1%, so a nuclear RTG becomes the smart choice. Radio signals take 3 to 22 minutes to reach Mars, and 79 minutes from Saturn.
>
> Our green lamps are inspired by JPL's design margins, and we label which thresholds are our own game rules. The storms come from stored NASA DONKI records, like May 2024."

The speaker for this slide studies section 3 and the "claims to avoid" list.

Slide 6: Demo video `2:30–3:15`

> "Let's see it fly." *[play the 40 s video and stay quiet]*

Slide 7: Impact, next steps, team `3:15–3:40`

> "AstroPixel is free and runs in any browser: no install, no account, no travel. We already put it in front of players, and the feedback was positive: they told us it's so detailed and teaches them so much

about missions and NASA data. Next, we'll test it with a school class here in Ramallah and add real NASA images.

Lose your first spacecraft here, for free, and learn why. Thank you."

3. Science study sheet (for slide 5 and the Q&A)

The golden rule: AstroPixel combines **physical relationships**, **NASA reference values**, **real hardware anchors** and **labelled game rules**. Say which one you mean. "We simplified that" plus a precise reason is a strong answer.

3.1 Mass vs weight

- `W = m · g` Mass stays the same everywhere; weight changes with local gravity.
- **Example:** a 1,000 kg craft weighs 9,810 N on Earth (g = 9.81), 1,620 N on the Moon (1.62), 3,730 N on Mars (3.73). Still 1,000 kg each time.
- Gravity values match NASA Planetary Fact Sheets (mean gravity; giant planets at the 1-bar level, not a surface).

3.2 The rocket equation: why mass snowballs

`Δv = v_e · ln(m₀ / m₁)` `v_e = Isp · g₀` `fuel = dry · (e^(Δv/v_e) − 1)`

- **Plain words:** fuel for later burns has to be pushed by earlier burns too, so fuel grows exponentially, not per kilometre.
- **Numbers:** Isp = 300 s (a configurable GMAT default, not a universal engine value) · g_0 = 9.81 · v_e = 2.943 km/s.
- **Worked Moon example:** direct plan Δv = 0.03 (course correction) + 0.90 (capture) + 2.00 (landing) = **2.93 km/s**. 1,000 kg dry → **1,706 kg fuel**, 2,706 kg wet. Adding the 3 kg camera adds **5.12 kg** of ideal fuel.
- **Reserve trap:** the 10% fuel reserve is 10% more *Δv* (3.223 km/s → 1,989.6 kg), not 10% more fuel (1,876.9 kg). Fuel is bought in 100 kg tanks, so it rounds up.
- **Mars with heat shield + parachute:** only 0.03 + 0.30 = 0.33 km/s of engine Δv (119 kg per 1,000 kg dry), because the shield and chute do the braking. That doesn't make entry easy; it is tested separately.

3.3 Launch: max-Q

- `q = ½ ρ v²` Dynamic pressure. Sea level ρ = 1.225 kg/m³: at 100 m/s, q = 6,125 Pa. Double the speed → four times q. That is why you throttle through max-Q.
- Our launch screen is a simplified model, not a manufacturer flight model.

3.4 Sunlight falls with distance squared

`P = P_1AU / r² · (1 − 0.05)^years · site factor`

- **Physical part:** inverse square. Mars (1.52 AU) gets **43%** of Earth's sunlight (586.2 vs 1,361 W/m², NASA Mars Fact Sheet). Saturn (9.54 AU) gets **1.1%**.
- **Model inputs:** 5% yearly array loss (a GMAT default) and a dust/site factor (Mars 0.8) are simplifications.
- **Example:** a 1,000 W array: 432.8 W at Mars before losses; 335 W after the 203-day cruise + 30 days with aging and dust. At Saturn after the 2,451-day cruise: about 7.75 W.
- **Say:** "solar is weak far out, so the RTG is the practical game option." **Don't say** solar is impossible at Saturn (Juno runs on solar at Jupiter).

3.5 RTGs and batteries

- **MMRTG (real hardware anchor, NASA/JPL):** about **110 W at the start of the mission**, 45 kg, at least 14 years expected operation, output declines gradually. It's a radioisotope generator, not a reactor.

- **Game rule:** output stays flat, then halves after 14 years. That step is a simplification, not real decay physics.
- `Wh = W × hours` Watts are a rate, watt-hours are stored energy. 100 W through the 12.3 h Mars night = 1,230 Wh; with the 35% allowance = 1,660.5 Wh, so one 2,000 Wh battery covers it.

3.6 Signal: delay, rate and volume are different things

`delay = distance / c` `Gb = Mb/s × seconds / 1000`

- **Delay:** Moon 1.28 s one way (384,400 km is the *mean* distance). Mars about **3–22 min** one way (NASA Mars Relay Network). Saturn at the game's 9.5 AU: about **79 min one way, 158 min round trip**.
- **Rate (game model):** antenna rate at 1 AU ÷ distance², with a cap. Example: our high-gain dish, 3 Mb/s at 1 AU → 1.33 Mb/s at Mars (1.5 AU); 8 contact hours a day for 30 days = 1,152 Gb.
- **Units:** Gb = gigabits, GB = gigabytes; 1 GB = 8 Gb.
- **Real anchors:** MRO's 3 m dish, *up to* 6 Mb/s from Mars. LRO's flown Ka-band: up to 100 Mb/s. Our 300 Mb/s cap is a game cap.
- **Honest:** the game shows the delay but your controls are not delayed by it.

3.7 Instruments: how one part couples to everything

- **Anchors (NASA pages):** MRO CTX camera 3 kg, 7 W · CRISM spectrometer ~33 kg, ~46 W, 2–2.5 Gb per typical targeted observation · Cassini MAG 3 kg, 3.1 W.
- **Game rules:** per-sample data sizes (camera 0.5 Gb, drill 0.1 Gb), prices and points.
- **Chain:** +1 camera → +3 kg and +7 W → with the 35% power allowance, +9.45 W of generation needed → +5.12 kg fuel on the Moon plan → more data to store and send. It *may* force another array, battery or antenna, which adds more mass and cost.

3.8 Margins: the denominator matters

Game check	Rule	What it means
Mass	(capacity − wet mass) / capacity ≥ 35%	wet mass ≤ 65% of the rocket's lift. 4,020 kg rocket → 2,613 kg
Power	generation ≥ 135% of demand	35% above demand
Battery	stored Wh ≥ 135% of night energy	
Data storage	storage ≥ 120% of expected data	16.67% empty at the threshold, not 20%
Comms	downlink ≥ 2 × science data	a game buffer, not a real 3 dB link budget
Budget	spending ≤ 75% of budget	a game convention; Discovery 2019's 25% reserve uses a different denominator

Source of the idea: JPL Concept Maturity Levels checklist (Wessen et al. 2013, p.13 Fig. 8b): ≥35% dry-mass, power and energy margin; ≥3 dB deep-space link. We **adapted** it. 3 dB ≈ a factor of 2 in *power*, which does not mean twice the data.

3.9 Trajectories and space weather

- **Paths:** stored durations and multipliers (some from real missions, some estimates). The game does **not** run GMAT, SPICE or live planet positions. Mars cruise 203 days is a stored reference, not an exact Hohmann solution.

- **Development time (game rule):** 200 days + 30 per new part type (+300 with nuclear power). 8 types, no RTG: 440 days; Mars total 440 + 203 + 30 = 673 days.
- **DONKI:** 15 stored 2024 NASA geomagnetic-storm (GST) records; every ID and peak Kp matches NASA. 10 May 2024 = Kp 9. Kp ≥ 7.5 → 40% base risk, otherwise 10% (**game odds**). Kp measures Earth's geomagnetic activity, not radiation at Mars or Saturn. Offline snapshot, not live.
- **Ring drone:** inspired by Cassini, not a reenactment. Cassini never launched a drone; its Grand Finale ended in Saturn's atmosphere.

3.10 Claims to avoid → say instead

Avoid	Say instead
"Every value is a real NASA number."	"We combine sourced reference values with scientific models and labelled game rules."
"NASA-approved simulation."	"An independent educational game using NASA and other primary sources."
"We use GMAT to calculate trajectories."	"We used GMAT defaults; Godot uses stored paths and our own calculations."
"NASA sends live storms into our game."	"The game includes a stored historical DONKI snapshot."
"Solar panels can't work at Saturn."	"They get about 1% of Earth's sunlight; our parts make the RTG the practical choice."
"RTGs give 110 W for 14 years, then lose half."	"About 110 W at the start; real output declines gradually. Our aging model is simplified."
"3 dB means twice the data."	"That margin idea inspired our separate 2× data buffer."
"Green lamps are NASA's exact requirements."	"JPL-inspired; each denominator is stated in the game."
"Cassini sent a drone into the rings."	"Our drone is a fictional mechanic inspired by Cassini's ring science."
"Robotic missions only."	"Robotic missions plus simplified crew tasks."

4. The hook: the three numbers

The team chose the stats as the hook. Deliver them on slide 2, one at a time, pointing at each panel:

> "149 million views of Artemis II. 406 million follows of NASA's accounts. 114,000 people signed up for Space Apps. The world loves space. But how many of those people have ever designed a mission themselves?"

Number	What it is exactly	Source
149 million	Views of NASA's Artemis II coverage across NASA platforms (views, not people)	NASA Artemis II streaming report, 2026
406.8 million	Account follows across NASA's social accounts (not unique people)	NASA 2024 year in review, "Reaching New, Future Explorers"
114,000+	Registered participants, Space Apps 2025 (114,094)	NASA 2025 Space Apps winners announcement and event metrics

These numbers show interest. They don't prove a user base for AstroPixel; don't claim they do.

5. Judges' questions: 3-beat answers (answer → evidence → impact, 20 s max)

Question	Answer
Impact: Who will use it, and how will it reach them?	Students and space clubs first, through teachers. It's one free web link, with no install and no account, so a school computer lab is enough. Next step: pilot with one Ramallah class.
Validity: Has anyone actually played it?	Yes. Playtest feedback was positive: players said it's very detailed and teaches a lot about missions and NASA data. *(Say how many people tested it and who they were: have the real number ready.)*
Creativity: What exists already?	Flight games are great at flying. Our focus is the whole mission chain: budget, power, data, comms and the team, with up to 4 roles and a report that explains every failure.
What NASA data did you actually use?	NASA Planetary Fact Sheet values (gravity, sunlight, distances), hardware anchors (MMRTG, MRO CTX/CRISM, Cassini MAG, MRO/LRO antennas), 15 stored DONKI storm records, GMAT documentation defaults, the JPL margin checklist. Some feed numbers, some inspire a rule.
Is it connected to live NASA services?	No. The storm records are an offline snapshot and the game computes everything locally. Multiplayer is a separate system.
Why does a small instrument need extra fuel?	For fixed Δv and Isp, fuel scales with dry mass through the rocket equation. On our Moon plan, 3 kg adds about 5.12 kg of fuel before reserve, plus power and data costs.
Does GMAT fly your mission?	No. We used GMAT documentation for defaults like 300 s Isp and 5%/year array loss. Paths come from stored reference tables.

Are the green lamps NASA's requirements?	No: JPL-inspired. Mass uses 35% unused lift, power 35% above demand, storage 20% above data. We state each denominator.
Is the storm damage chance a NASA measurement?	No. Real DONKI dates and Kp, then our own odds (40% strong / 10% otherwise). Kp is an Earth index, not a dose at the spacecraft.
What's simplified, and why is it still educational?	Fixed distances, stored paths, simplified link and energy models, tuned hazards. The value is learning coupled trade-offs and telling evidence from assumptions.
Is it only for experts?	No. EASY mode has a Mission Control guide and auto-build, and the science is explained in plain words inside the game.
Did you use AI?	Yes, for code help and drafting text. The idea and the game design are ours. No AI images, video or audio: the art comes from our own Blender and Python scripts.

6. Fact-check before the pitch

- **Palestine 1:** "no aerospace engineering degree we could find". Check the programme lists (An-Najah, Birzeit, IUG, Al-Quds, PTUK). Say "we found", not "there is none".
- **Palestine 2:** the Smithsonian National Air and Space Museum × Al Nayzak programme takes Palestinian teens (14–17) to NASA Goddard (global.si.edu).
- **Palestine 3:** volunteer astronomy clubs (e.g. NOVA in Ramallah). Our pitch adds to them; don't say there is nothing.
- **Playtest:** have the real number of testers ready.
- **Live demo:** open the web game and relay server about 2 minutes before you pitch (the free server needs about a minute to wake up). Keep the video offline on the laptop and a USB stick.

Science sources (from the team's audit): NASA Glenn (Ideal Rocket Equation, Dynamic Pressure) · NASA Planetary Fact Sheets · NASA/JPL Perseverance press kit, Power (MMRTG) · NASA Mars Relay Network (3–22 min) · NASA MRO and Cassini instrument pages · JPL IPN 42-222B (LRO 100 Mb/s) · JPL CML paper, p.13 Fig. 8b · Discovery 2019 AO §5.6.3 · NASA GMAT source (Isp 300 s, 5%/yr) · NASA CCMC DONKI · NOAA Kp explanation. Palestine: Smithsonian Global · EduRank · Astrobites.